

$$4.3 \quad G_0 = B_0 \oplus B_1 \quad G_1 = B_1 \oplus B_2 \quad G_2 = B_2 \oplus B_3 \quad G_3 = B_3$$

$B_3 B_2 B_1 B_0$	G_3	G_2	G_1	G_0
0 0 0 0	0	0	0	0
0 0 0 1	0	0	0	1
0 0 1 0	0	0	1	1
0 0 1 1	0	0	1	0
0 1 0 0	0	1	1	0
0 1 0 1	0	1	1	1
0 1 1 0	0	1	0	1
0 1 1 1	0	1	0	0
1 0 0 0	1	1	0	0
1 0 0 1	1	1	0	1
1 0 1 0	1	1	1	1
1 0 1 1	1	1	1	0
1 1 0 1	1	0	1	1

1 1 1 0 1 0 0 1
 1 1 1 1 1 0 0 0

8421 码转经典格雷码

4.5 解:

$$K=1 \text{ 时, } A_2 = B_2, \quad A_1 = B_2 \oplus B_1, \quad A_0 = B_0 \oplus B_1$$

$$K=0 \text{ 时, } A_2 = B_2, \quad A_1 = B_2 \oplus B_1, \quad A_0 = B_0 \oplus B_1 \oplus B_2$$

总逻辑表达式 $A_2 = B_2, \quad A_1 = B_2 \oplus B_1$

$$A_0 = [(B_1 \oplus B_2) \bar{K} + KB_1] \oplus B_0$$

$K=1$ 时:

B_2	B_1	B_0	A_2	A_1	A_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	1
1	1	1	1	0	0

$K=0$ 时:

B_2	B_1	B_0	A_2	A_1	A_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	0	0	1	1
0	1	1	0	1	0
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	1	0	0
1	1	1	1	0	1

三位二进制转格雷码

格雷码转三位二进制

$$4.6. \quad A = \overline{I_1 I_3 I_7 I_5 I_9} = I_1 + I_3 + I_5 + I_7 + I_9$$

$$B = \overline{I_2 I_6 I_4 I_7} = I_2 + I_6 + I_4 + I_7$$

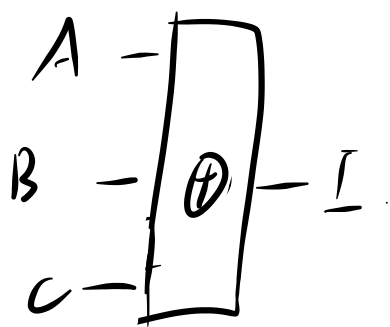
$$C = \overline{I_4 I_6 I_5 I_7} = I_4 + I_5 + I_6 + I_7$$

$$D = \overline{I_8 I_9} = I_8 + I_9$$

8421 码的编码表

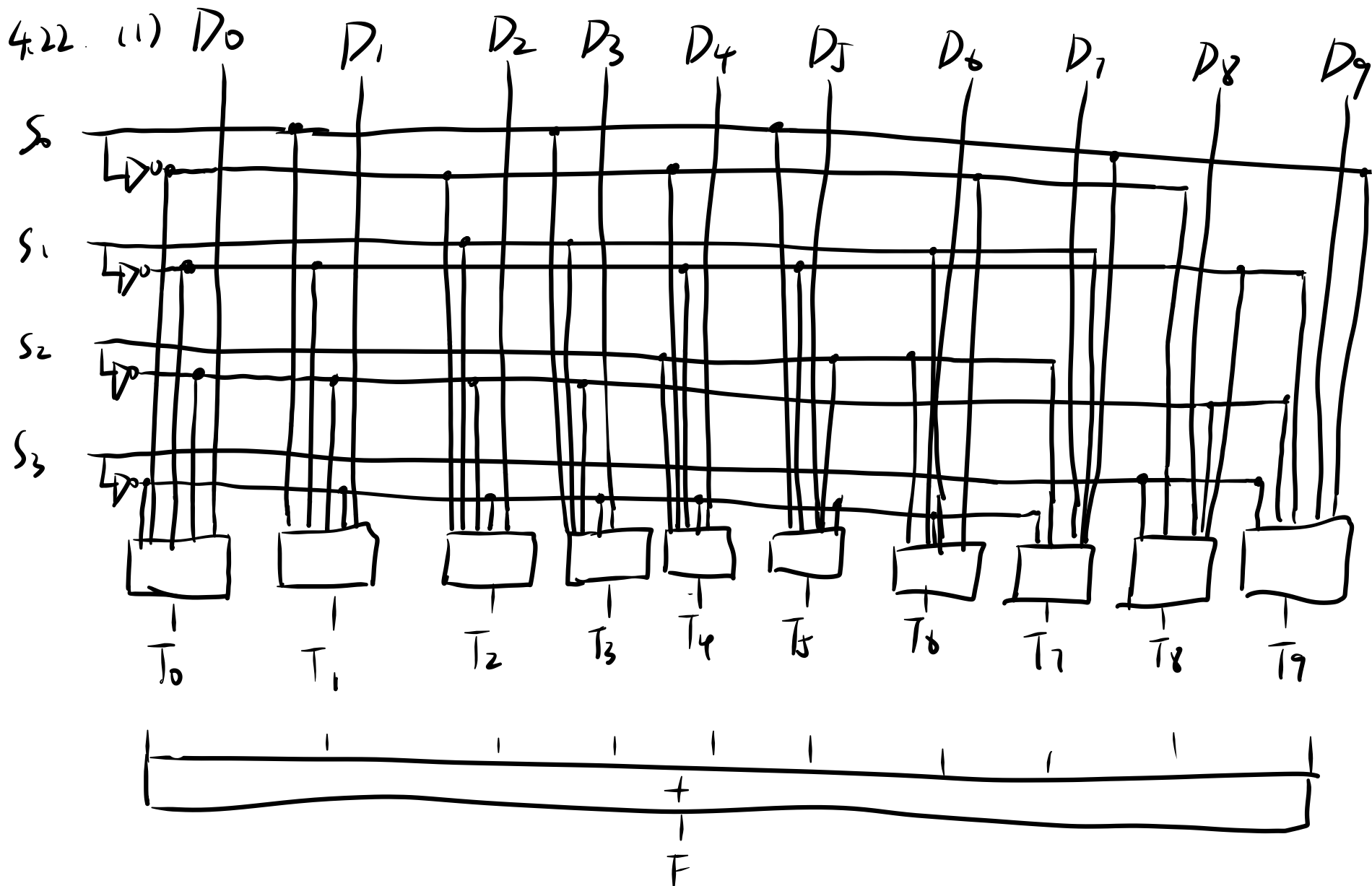
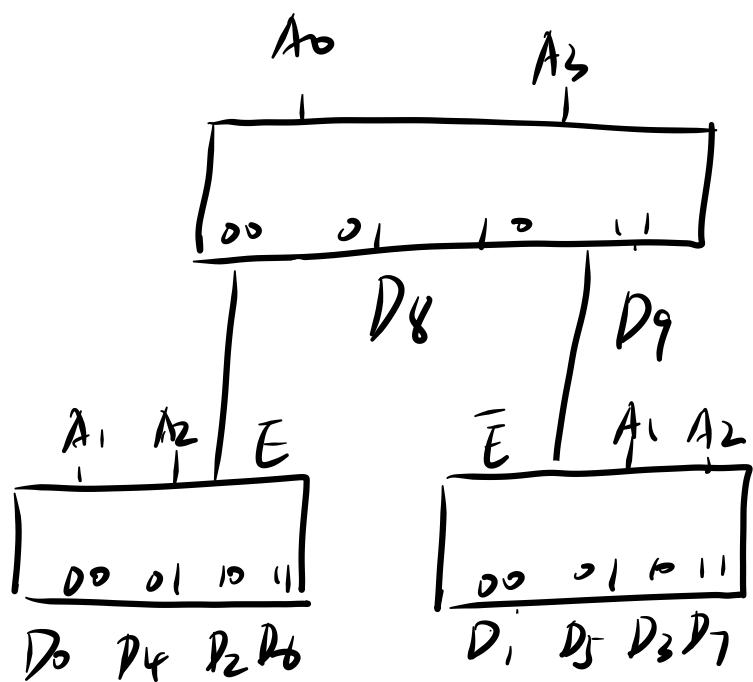
4.10 输出状态改变则输出改变

ABC	L
000	0
001	1
010	1
100	1
011	0
110	0
101	0
111	1



$$L = A \oplus B \oplus C$$

4.15



12) - 块 8 进 - + - 块 4 进 = 1# -

